EXPT NO:5	DEVELOP INTERACTIVE DASHBOARDS FOR DYNAMIC DATA EXPLORATION USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. **What is dynamic data exploration in BI tools, and why is it important?**
   Dynamic data exploration is the process of interactively analyzing data using filters, slicers, drill-downs, and other visual tools. It allows users to view data from different perspectives without changing the original dataset. It is important because it helps identify trends, patterns, and insights quickly, leading to better and faster decision-making.

2. **What are slicers and filters, and how do they enhance dashboard interactivity?**
   - Slicers and filters allow users to display only the data they need.
   - Filters limit the data shown in charts and tables.
   - Slicers are visual filter controls that users can click to change dashboard views.
   - They improve dashboard interactivity by enabling users to analyze specific categories, regions, products, or time periods instantly.

3. **How do Tableau and Power BI support drill-down and drill-through features?**
   Both Tableau and Power BI allow users to move from summarized information to detailed data.
   - **Drill-down** lets users navigate through hierarchical levels, such as Year → Quarter → Month → Day.
   - **Drill-through** (Power BI) allows users to open a detailed report page related to the selected data.

   These features help users perform deeper analysis without creating separate reports.

4. **What is the role of parameters in creating interactive dashboards?**
   Parameters allow users to change values dynamically within a dashboard. They can be used to switch between different KPIs, adjust calculations, or control visualizations. Parameters make dashboards more flexible and interactive.

5. **Name some advanced interactivity features in Tableau and Power BI.**
   Some advanced interactivity features include:
   - Filters and Slicers
   - Parameters
   - Drill-down
   - Drill-through
   - Tooltips
   - Highlight Actions
   - Bookmarks (Power BI)
   - Dashboard Actions
   - Cross-filtering
   - Cross-highlighting

IN-LAB EXERCISE:

OBJECTIVE:

To design and develop interactive dashboards in Tableau and Power BI for dynamic data exploration, enabling users to analyze, filter, and drill down into data for deeper insights.

PROCEDURE:

1. **Data Collection:**
    - Import datasets from sources such as Excel, CSV, SQL Server, or online APIs.
    - Dataset: *Retail Sales Dataset* with fields such as Order Date, Region, Product Category, Sales, Profit, Quantity.
2. **Data Preparation and Transformation:**
    - Ensure proper date formats for time-based analysis.
    - Handle missing or incorrect values.
    - Create calculated fields (e.g., Profit Margin %, Sales Growth %, Year-to-Date Sales).
3. **Data Aggregation and Hierarchies:**
    - Create hierarchies (e.g., Year → Quarter → Month → Day).
    - Aggregate data by region, product category, or customer segment.
4. **Designing Interactive Elements:**
    - **Filters & Slicers:** Allow filtering by product category, date range, or region.
    - **Parameters:** Let users switch between KPIs (e.g., Sales vs. Profit).
    - **Drill-down:** Enable users to explore data from summary to detail.
    - **Tooltips:** Add contextual details when hovering over data points.
5. **Dashboard Development:**
    - **Tableau:** Use interactive filters, parameter controls, and highlight actions.
    - **Power BI:** Use slicers, bookmarks, and drill-through pages.
    - Add KPI cards for key metrics like *Total Sales*, *Total Profit*, *Top Selling Product*.
    - Use maps for regional sales analysis, line charts for trends, and bar charts for category comparisons.
6. **Testing Interactivity:**
    - Ensure slicers and filters update all relevant visuals dynamically.
    - Test drill-down and drill-through functionality for accuracy.
7. **Publishing and Sharing:**

- **Tableau:** Publish on Tableau Public or Tableau Server.
- **Power BI:** Publish to Power BI Service and share via workspace access or public link.

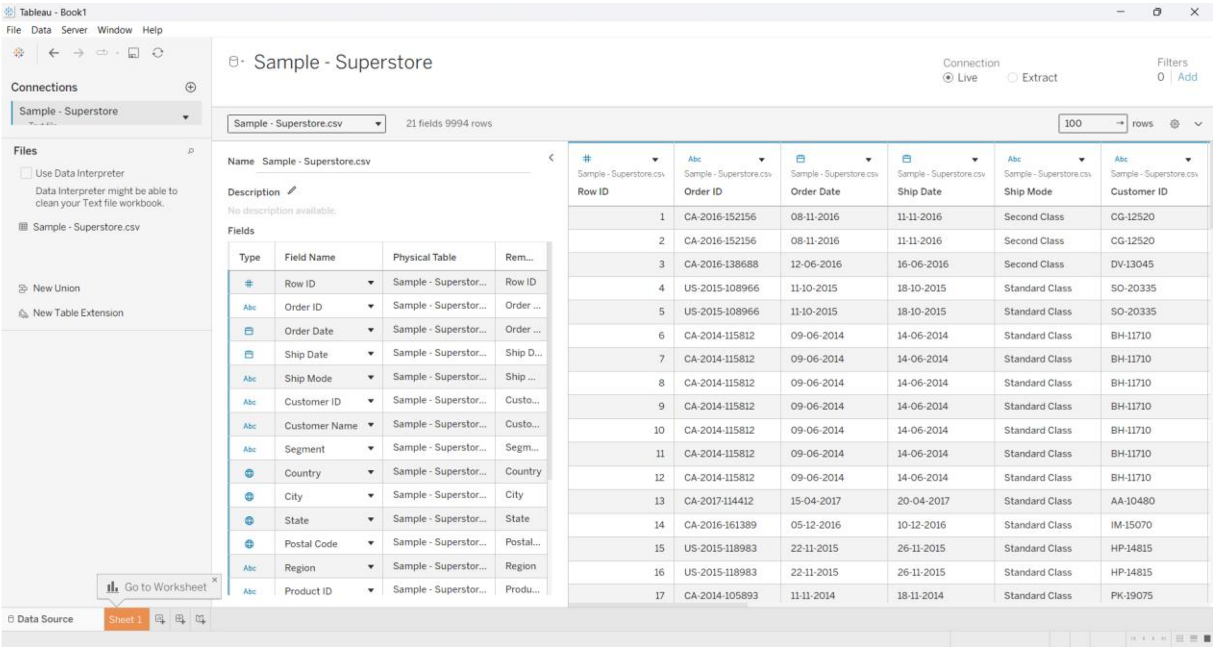
Experiment Dataset:

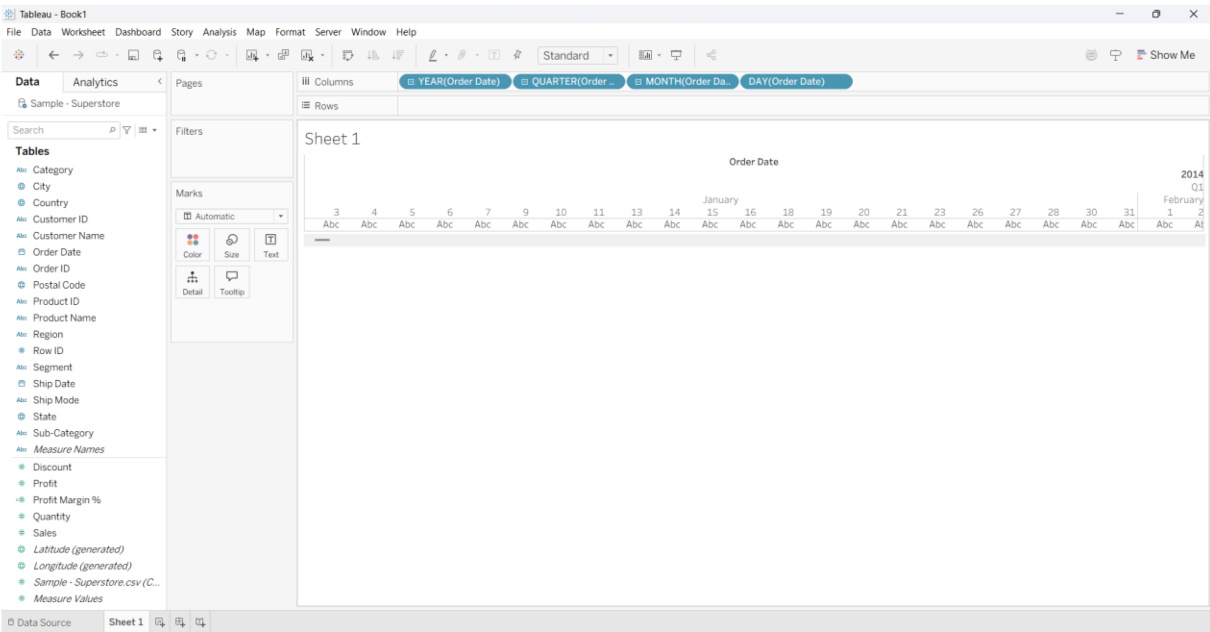
Retail Sales Dataset

Fields Used:

- **Order Date – Date of the order**
- **Region – Sales region**
- **Product Category – Category of product sold**
- **Sales – Sales revenue**
- **Profit – Net profit from the order**
- **Quantity – Units sold**

TABLEAU STEPS :





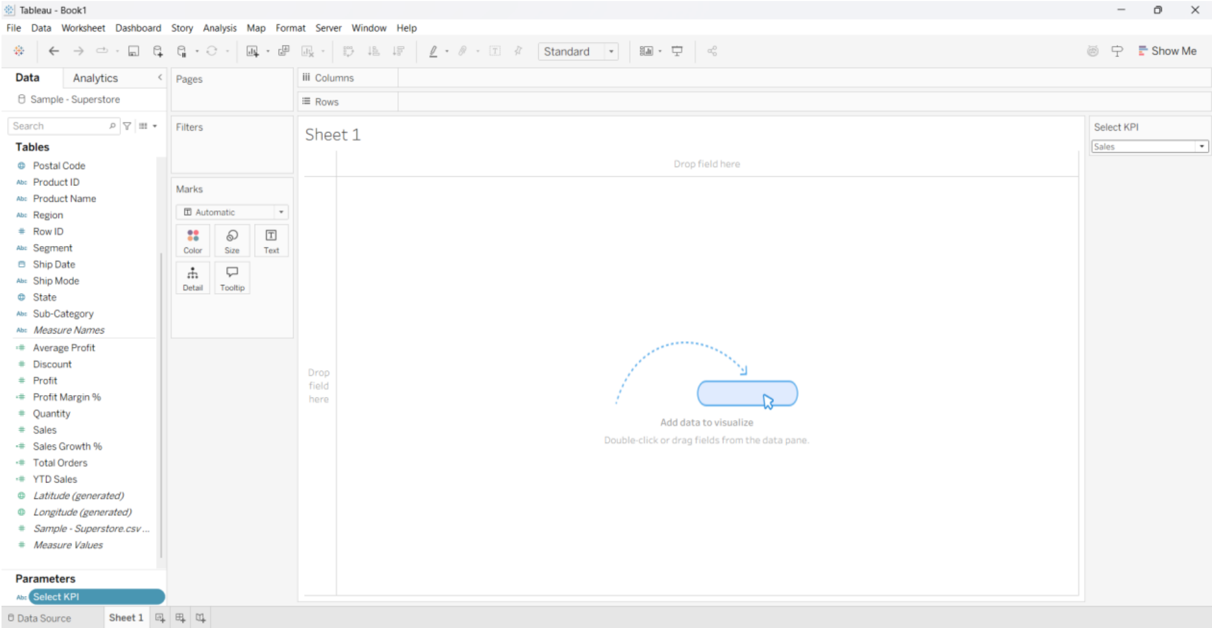


Tableau - Book1
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Data Analytics
Sample - Superstore
Pages
Columns
Rows
Filters
Marks
Sheet 1
Drop field here
Select KPI
Sales
Tables
Postal Code
Product ID
Product Name
Region
Row ID
Segment
Ship Date
Ship Mode
State
Sub-Category
Measure Names
Average Profit
Discount
Profit
Profit Margin %
Quantity
Sales
Sales Growth %
Total Orders
YTD Sales
Latitude (generated)
Longitude (generated)
Measure Values
Parameters
Select KPI
Add data to visualize
Double-click or drag fields from the data pane.
Data Source
Sheet 1

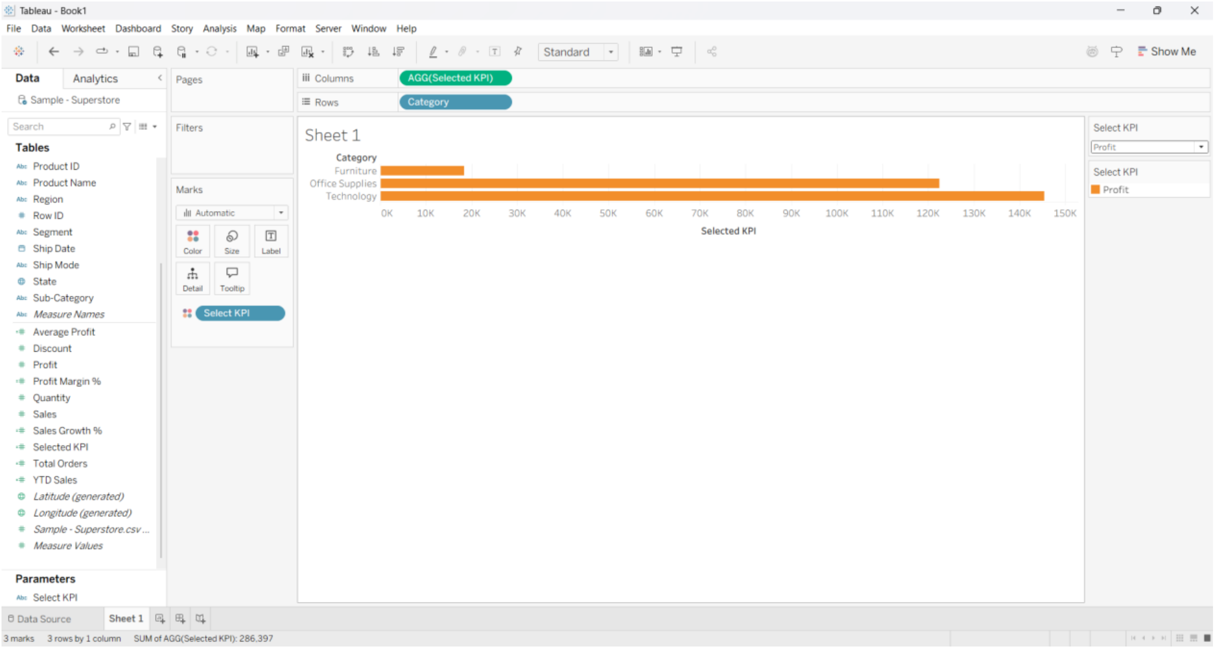


Tableau - Book1
Columns AGG(Selected KPI)
Rows Category
Sheet 1
Category
Furniture
Office Supplies
Technology
0K 10K 20K 30K 40K 50K 60K 70K 80K 90K 100K 110K 120K 130K 140K 150K
Selected KPI
Select KPI
Profit
Select KPI
Profit
Marks
Automatic
Select KPI
Selected KPI
Parameters
Select KPI
3 marks 3 rows by 1 column SUM of AGG(Selected KPI): 286,397

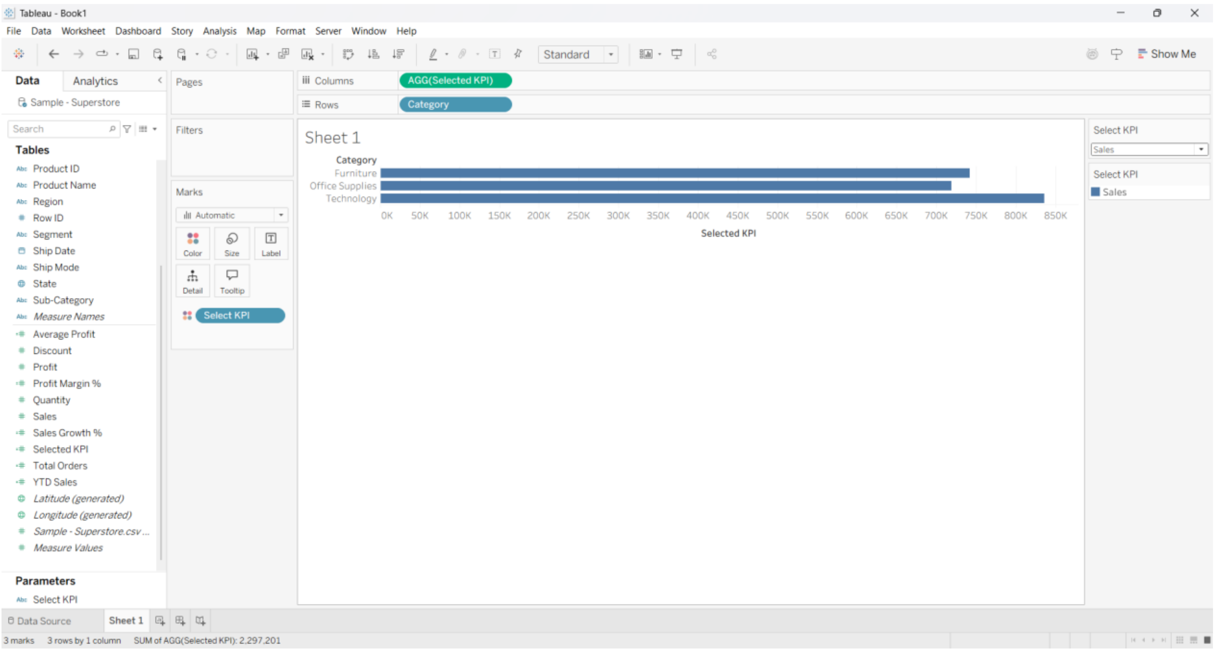
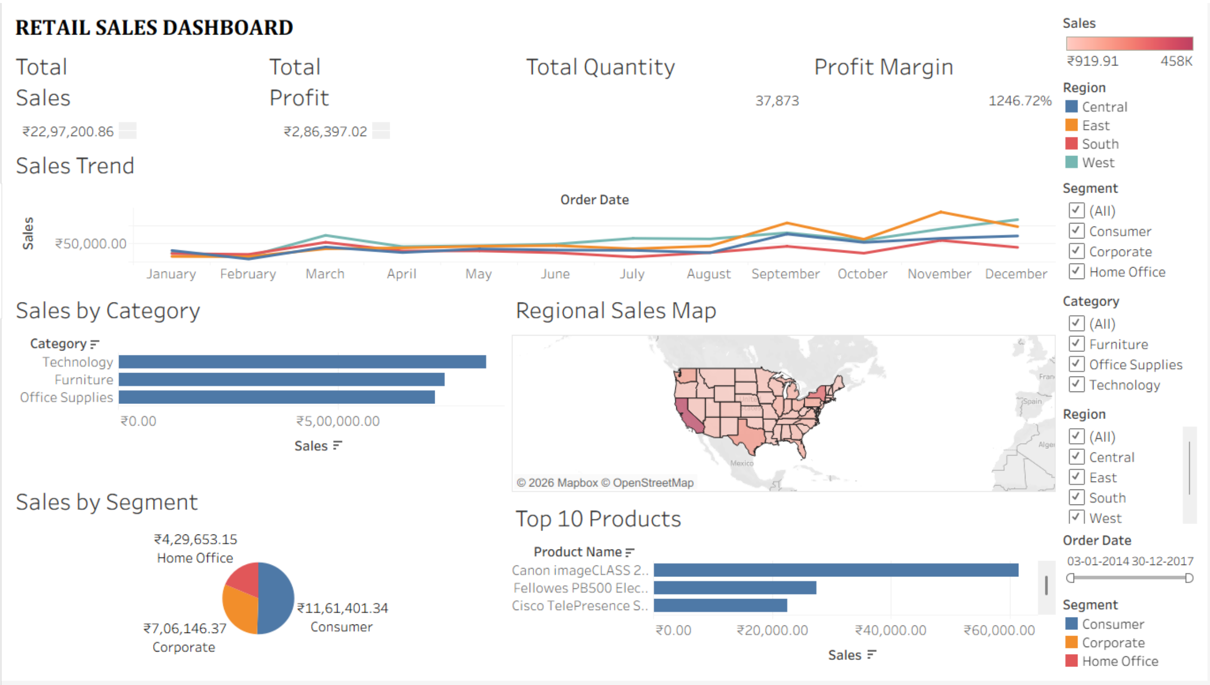
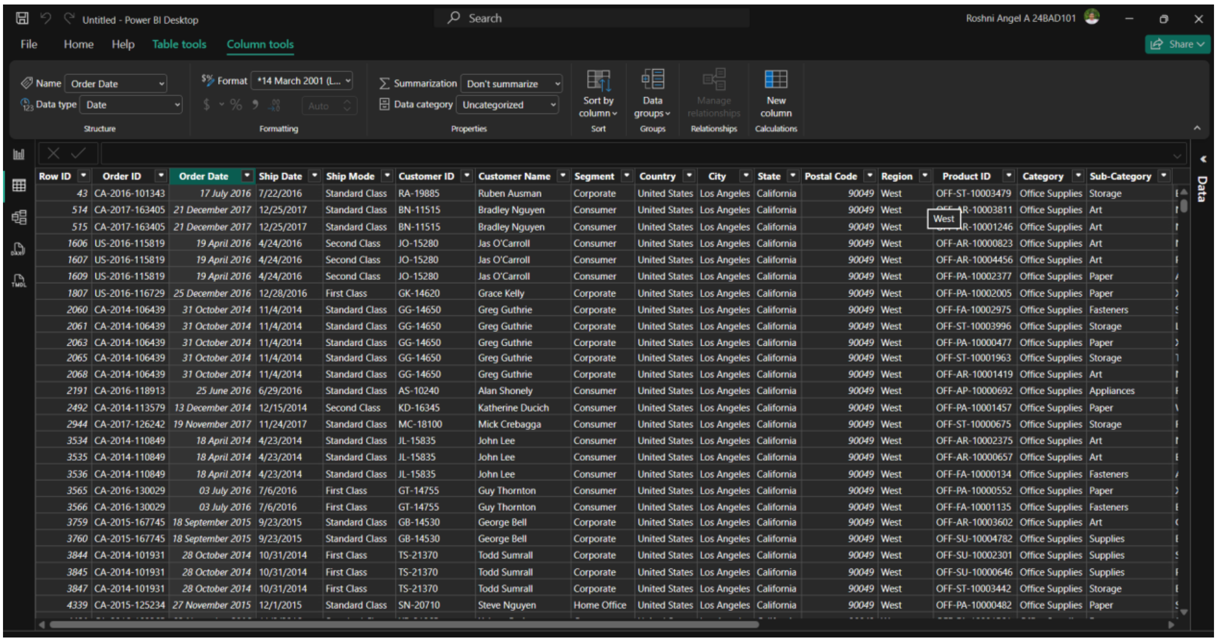


Tableau - Book1
Columns AGG(Selected KPI)
Rows Category
Sheet 1
Category
Furniture
Office Supplies
Technology
0K 50K 100K 150K 200K 250K 300K 350K 400K 450K 500K 550K 600K 650K 700K 750K 800K 850K
Selected KPI
Select KPI
Sales
Select KPI
Sales
Marks
Automatic
Select KPI
Selected KPI
Parameters
Select KPI
3 marks 3 rows by 1 column SUM of AGG(Selected KPI): 2,297,201

FINAL TABLEAU DASHBOARD :



POWER BI STEPS :



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
43	CA-2016-101343	17 July 2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003479	Office Supplies	Storage
514	CA-2017-163405	21 December 2017	12/25/2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California	90049	West	[illegible]R-10003811	Office Supplies	Art
515	CA-2017-163405	21 December 2017	12/25/2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California	90049	West	[illegible]R-10001246	Office Supplies	Art
1606	US-2016-115819	19 April 2016	4/24/2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10000823	Office Supplies	Art
1607	US-2016-115819	19 April 2016	4/24/2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10004456	Office Supplies	Art
1609	US-2016-115819	19 April 2016	4/24/2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10002377	Office Supplies	Paper
1807	US-2016-116729	25 December 2016	12/28/2016	First Class	GK-14620	Grace Kelly	Corporate	United States	Los Angeles	California	90049	West	OFF-PA-10002005	Office Supplies	Paper
2060	CA-2014-106439	31 October 2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-FA-10002975	Office Supplies	Fasteners
2061	CA-2014-106439	31 October 2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003996	Office Supplies	Storage
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2065	CA-2014-106439	31 October 2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10001963	Office Supplies	Storage
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2191	CA-2016-118913	25 June 2016	6/29/2016	Standard Class	AS-10240	Alan Shonely	Consumer	United States	Los Angeles	California	90049	West	OFF-AP-10000692	Office Supplies	Appliances
2492	CA-2014-113579	13 December 2014	12/15/2014	Second Class	KD-16345	Katherine Ducich	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10001457	Office Supplies	Paper
2944	CA-2017-126242	19 November 2017	11/24/2017	Standard Class	MC-18100	Mick Crebagga	Consumer	United States	Los Angeles	California	90049	West	OFF-ST-10000675	Office Supplies	Storage
3534	CA-2014-110849	18 April 2014	4/23/2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10002375	Office Supplies	Art
3535	CA-2014-110849	18 April 2014	4/23/2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10000657	Office Supplies	Art
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3565	CA-2016-130029	03 July 2016	7/6/2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10000552	Office Supplies	Paper
3566	CA-2016-130029	03 July 2016	7/6/2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California	90049	West	OFF-FA-10001135	Office Supplies	Fasteners
3759	CA-2015-167745	18 September 2015	9/23/2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California	90049	West	OFF-AR-10003602	Office Supplies	Art
3760	CA-2015-167745	18 September 2015	9/23/2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California	90049	West	OFF-SU-10004782	Office Supplies	Supplies
3844	CA-2014-101931	28 October 2014	10/31/2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West	OFF-SU-10002301	Office Supplies	Supplies
3845	CA-2014-101931	28 October 2014	10/31/2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West	OFF-SU-10000646	Office Supplies	Supplies
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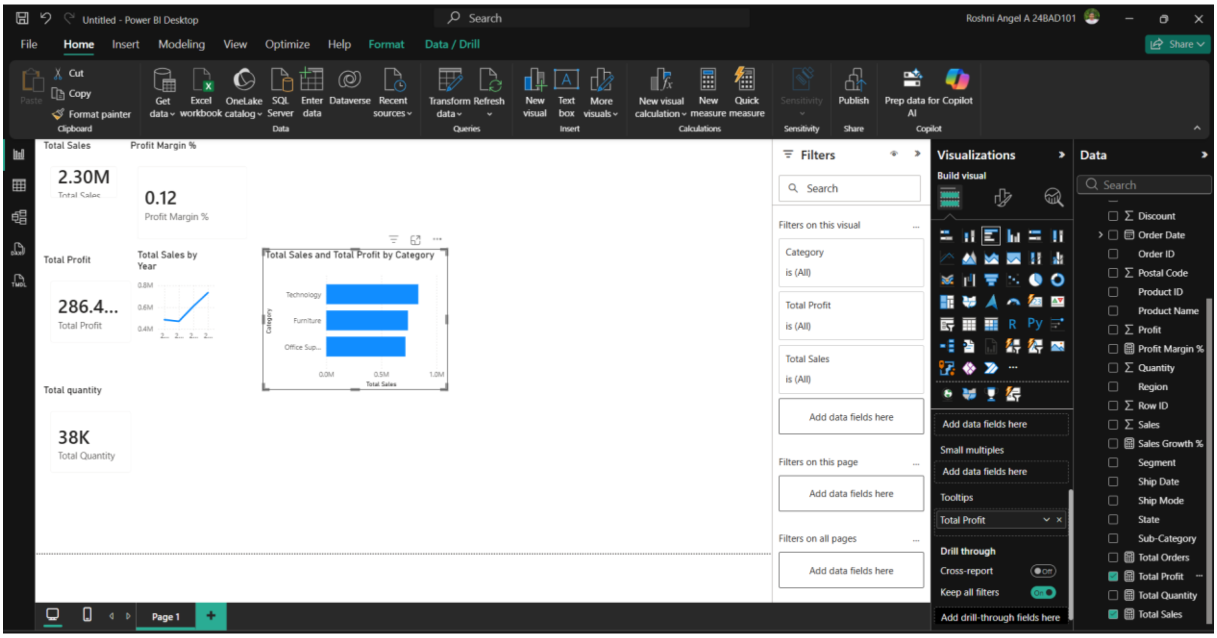
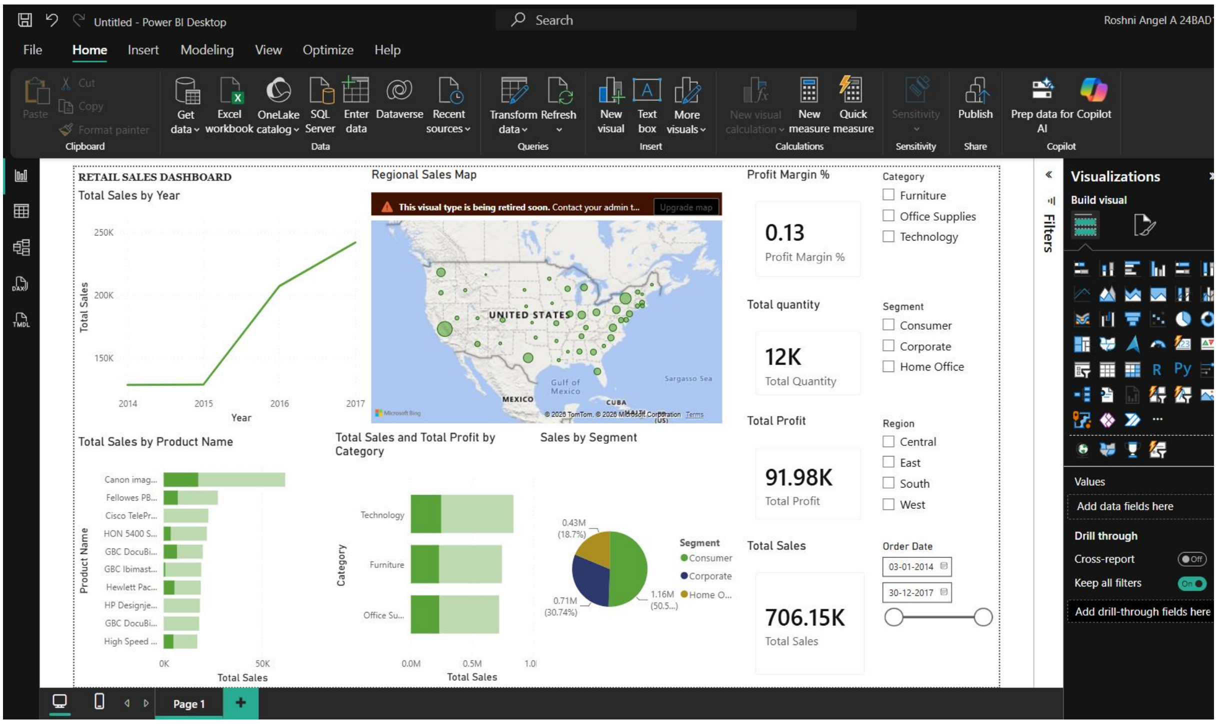
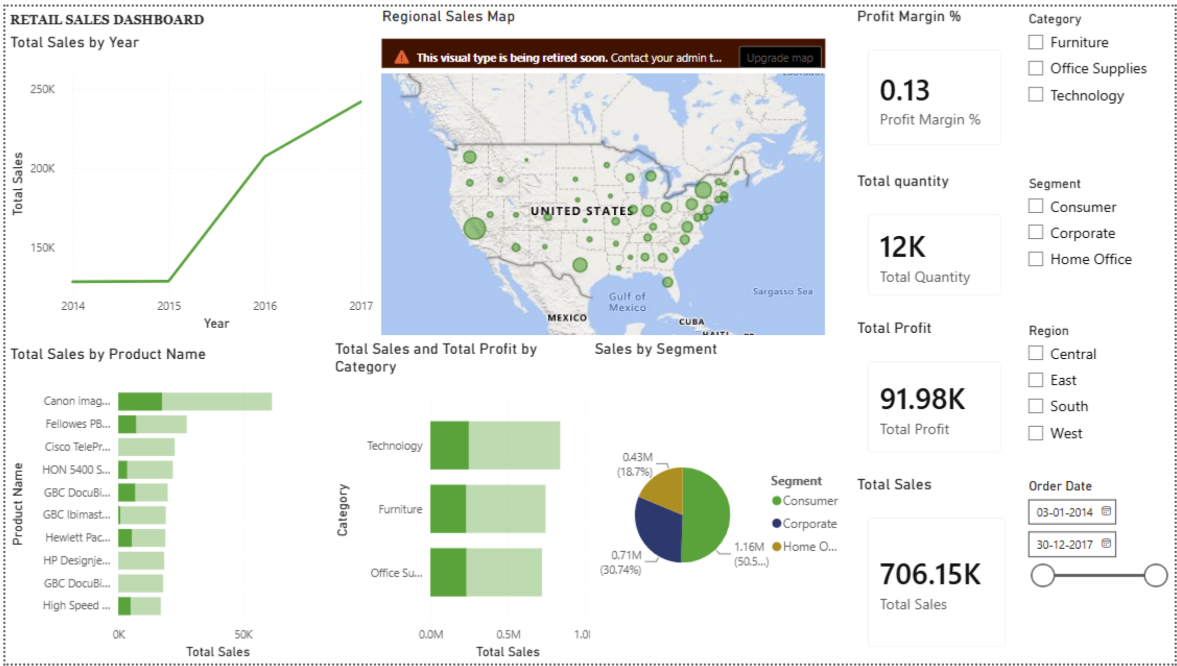
4339	CA-2015-125234	27 November 2015	12/1/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Los Angeles	California	90049	West	OFF-PA-10000482	Office Supplies	Paper



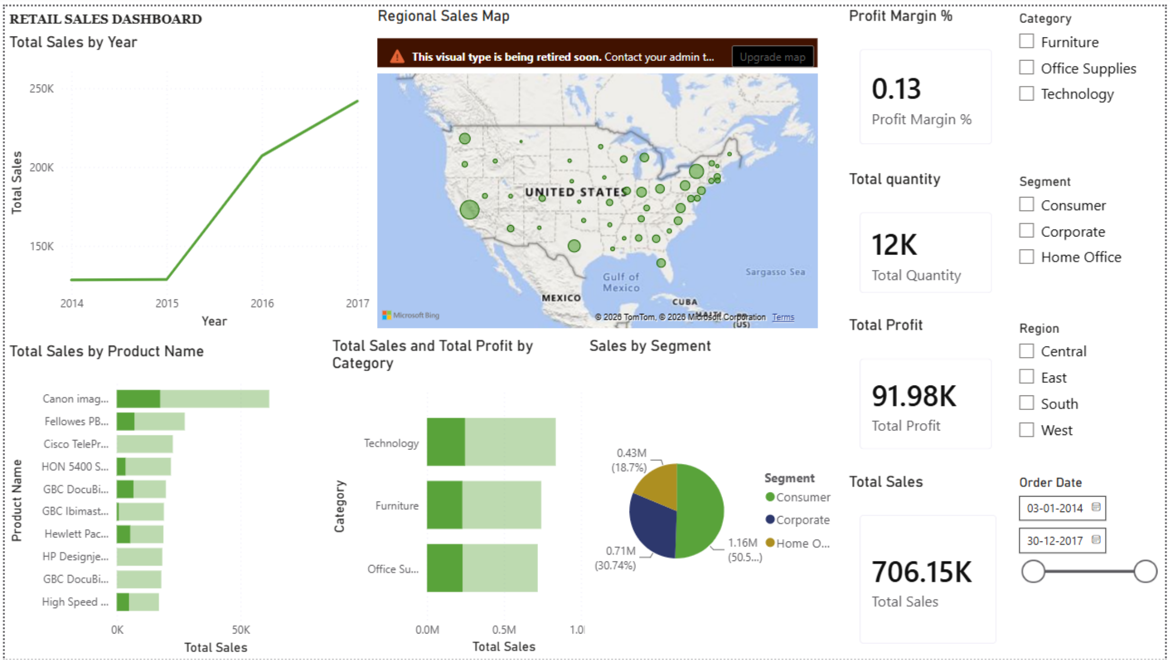
Untitled - Power BI Desktop
Search
Roshni Angel A 24BAD101
File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Total Sales
2.30M
Profit Margin %
0.12
Total Profit
286.4...
Total Sales by Year
Total Sales and Total Profit by Category
Technology
Furniture
Office Sup...
Total quantity
38K
Total Quantity
Filters
Filters on this visual
Category
is (All)
Total Profit
is (All)
Total Sales
is (All)
Add data fields here
Filters on this page
Filters on all pages
Visualizations
Build visual
Small multiples
Tooltips
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Discount
Order Date
Order ID
Postal Code
Product ID
Product Name
Profit
Profit Margin %
Quantity
Region
Row ID
Sales
Sales Growth %
Segment
Ship Date
Ship Mode
State
Sub-Category
Total Orders
Total Profit
Total Quantity
Total Sales
Page 1



Untitled - Power BI Desktop
Search
File
Home
Insert
Modeling
View
Optimize
Help
RETAIL SALES DASHBOARD
Total Sales by Year
250K
200K
150K
2014
2015
2016
2017
Year
Total Sales
Regional Sales Map
This visual type is being retired soon. Contact your admin t...
Upgrade map
UNITED STATES
MEXICO
Gulf of Mexico
Sargasso Sea
CUBA
Profit Margin %
0.13
Profit Margin %
Total quantity
12K
Total Quantity
Total Profit
91.98K
Total Profit
Total Sales
706.15K
Total Sales
Category
Furniture
Office Supplies
Technology
Segment
Consumer
Corporate
Home Office
Region
Central
East
South
West
Order Date
03-01-2014
30-12-2017
Total Sales by Product Name
Canon imag...
Fellowes PB...
Cisco TelePr...
HON 5400 S...
GBC DocuBi...
GBC Ibimast...
Hewlett Pac...
HP Designje...
GBC DocuBi...
High Speed ...
Product Name
0K
50K
Total Sales
Total Sales and Total Profit by Category
Technology
Furniture
Office Su...
Category
0.0M
0.5M
Total Sales
Sales by Segment
0.43M
(18.7%)
0.71M
(30.74%)
1.16M
(50.5...)
Segment
Consumer
Corporate
Home O...
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Filters
Page 1



Final Power BI dashboard :



Power BI link :

https://kumaragurudtsteam-my.sharepoint.com/:u:/r/personal/roshniangel_24ad_kct_ac_in/Documents/English/Retail%20power%20BI%20dashboard.pbix?csf=1&web=1&e=aTgrAX

POST-LAB QUESTIONS

1. **How did interactivity improve the analysis experience in your dashboard?**

   Interactivity made the dashboard more user-friendly by allowing users to explore the data dynamically. Features such as filters, slicers, drill-down, and tooltips enabled quick analysis of sales, profit, quantity, and regional performance without modifying the original dataset. This improved decision-making by providing instant insights based on user selections.

2. **Which interactive feature (filters, drill-down, parameters, etc.) was most useful and why?**

   The most useful interactive feature was the **filters and slicers** because they allowed the dashboard to display specific information based on selected regions, product categories, segments, and dates. This made it easier to compare performance across different categories and identify trends quickly. Drill-down was also helpful for analyzing data from yearly summaries to monthly details.

3. **What challenges did you face while implementing drill-down or dynamic filters?**

   The main challenges included configuring the date hierarchy correctly for drill-down, ensuring slicers updated all visuals simultaneously, and creating calculated measures for KPIs. Some visual formatting and hierarchy settings also required additional adjustments to display the expected results correctly.

4. **How do tooltips contribute to better dashboard storytelling?**

   Tooltips provide additional information when users hover over a visual without overcrowding the dashboard. They display detailed values such as sales, profit, and quantity, helping users understand the context behind each data point. This makes the dashboard more informative, interactive, and easier to interpret.

5. **What additional features would you add to make the dashboard more insightful?**

   To make the dashboard more insightful, I would add:
   - Forecasting for future sales trends.
   - Conditional formatting to highlight high and low performance.
   - Advanced KPI indicators with target comparisons.
   - Customer segmentation analysis.
   - Dynamic bookmarks and navigation buttons.
   - Drill-through pages for detailed product and regional analysis.
   - AI-based visuals such as Key Influencers and Decomposition Tree in Power BI.
   - Automated alerts for significant changes in sales or profit.

   These features would enhance the dashboard by providing deeper insights and supporting better business decision-making.

RESULT:

The retail sales dataset was successfully transformed, aggregated, and visualized into interactive dashboards in Tableau and Power BI. Filters, slicers, drill-downs, and parameters enabled dynamic exploration of sales and profit trends by region, product category, and time period. The resulting dashboards allowed for quick insights and deeper analysis, enhancing decision-making capabilities.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	